

Lab - Accessing Network Devices with SSH (Instructor Version – Optional Lab)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only. Optional activities are designed to enhance understanding and/or to provide additional practice.

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.11	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Objectives

Part 1: Configure Basic Device Settings

Part 2: Configure the Router for SSH Access

Part 3: Configure the Switch for SSH Access

Part 4: SSH from the CLI on the Switch

Background / Scenario

In the past, Telnet was the most common network protocol used to remotely configure network devices. Telnet does not encrypt the information between the client and server. This allows a network sniffer to intercept passwords and configuration information.

Secure Shell (SSH) is a network protocol that establishes a secure terminal emulation connection to a router or other networking device. SSH encrypts all information that passes over the network link and provides authentication of the remote computer. SSH is rapidly replacing Telnet as the remote login tool of choice for network professionals. SSH is most often used to log in to a remote device and execute commands; however, it can also transfer files using the associated Secure FTP (SFTP) or Secure Copy (SCP) protocols.

The network devices that are communicating must be configured to support SSH in order for SSH to function. In this lab, you will enable the SSH server on a router and then connect to that router using a PC with an SSH client installed. On a local network, the connection is normally made using Ethernet and IP.

Note: The routers used with CCNA hands-on labs are Cisco 1941 Integrated Services Routers (ISRs) with Cisco IOS Release 15.2(4)M3 (universalk9 image). The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.0(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of this lab for the correct interface identifiers.

Note: Make sure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Instructor Note: Refer to the Instructor Lab Manual for the procedures to initialize and reload devices.

Required Resources

- 1 Router (Cisco 1941 with Cisco IOS Release 15.2(4)M3 universal image or comparable)
- 1 Switch (Cisco 2960 with Cisco IOS Release 15.0(2) lanbasek9 image or comparable)
- 1 PC (Windows 7 or 8 with terminal emulation program, such as Tera Term, and Wireshark installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Part 1: Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings, such as the interface IP addresses, device access, and passwords on the router.

Step 1: Cable the network as shown in the topology.

Step 2: Initialize and reload the router and switch.

Step 3: Configure the router.

- Console into the router and enable privileged EXEC mode.
- Enter configuration mode.
- Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- Assign **class** as the privileged EXEC encrypted password.
- Assign **cisco** as the console password and enable login.
- Assign **cisco** as the VTY password and enable login.
- Encrypt the plaintext passwords.
- Create a banner that will warn anyone accessing the device that unauthorized access is prohibited.
- Configure and activate the G0/1 interface on the router using the information contained in the Addressing Table.
- Save the running configuration to the startup configuration file.

Step 4: Configure PC-A.

- Configure PC-A with an IP address and subnet mask.
- Configure a default gateway for PC-A.

Step 5: Verify network connectivity.

Ping R1 from PC-A. If the ping fails, troubleshoot the connection.

Part 2: Configure the Router for SSH Access

Using Telnet to connect to a network device is a security risk because all information is transmitted in a clear text format. SSH encrypts the session data and provides device authentication, which is why SSH is recommended for remote connections. In Part 2, you will configure the router to accept SSH connections over the VTY lines.

Step 1: Configure device authentication.

The device name and domain are used as part of the crypto key when it is generated. Therefore, these names must be entered prior to issuing the **crypto key** command.

- a. Configure device name.

```
Router(config)# hostname R1
```

- b. Configure the domain for the device.

```
R1(config)# ip domain-name ccna-lab.com
```

Step 2: Configure the encryption key method.

```
R1(config)# crypto key generate rsa modulus 1024
```

The name for the keys will be: R1.ccna-lab.com

% The key modulus size is 1024 bits

% Generating 1024 bit RSA keys, keys will be non-exportable...

[OK] (elapsed time was 1 seconds)

```
R1(config)#
```

*Jan 28 21:09:29.867: %SSH-5-ENABLED: SSH 1.99 has been enabled

Step 3: Configure a local database username.

```
R1(config)# username admin privilege 15 secret adminpass
```

Note: A privilege level of 15 gives the user administrator rights.

Step 4: Enable SSH on the VTY lines.

- a. Enable Telnet and SSH on the inbound VTY lines using the **transport input** command.

```
R1(config)# line vty 0 4
```

```
R1(config-line)# transport input telnet ssh
```

- b. Change the login method to use the local database for user verification.

```
R1(config-line)# login local
```

```
R1(config-line)# end
```

```
R1#
```

Step 5: Save the running configuration to the startup configuration file.

```
R1# copy running-config startup-config
```

Destination filename [startup-config]?

Building configuration...

[OK]

```
R1#
```

Step 6: Establish an SSH connection to the router.

- a. Start Tera Term from PC-A.
- b. Establish an SSH session to R1. Use the username **admin** and password **adminpass**. You should be able to establish an SSH session with R1.

Part 3: Configure the Switch for SSH Access

In Part 3, you will configure the switch in the topology to accept SSH connections. After the switch has been configured, establish an SSH session using Tera Term.

Step 1: Configure the basic settings on the switch.

- a. Console into the switch and enable privileged EXEC mode.
- b. Enter configuration mode.
- c. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- d. Assign **class** as the privileged EXEC encrypted password.
- e. Assign **cisco** as the console password and enable login.
- f. Assign **cisco** as the VTY password and enable login.
- g. Encrypt the plain text passwords.
- h. Create a banner that will warn anyone accessing the device that unauthorized access is prohibited.
- i. Configure and activate the VLAN 1 interface on the switch according to the Addressing Table.
- j. Save the running configuration to the startup configuration file.

Step 2: Configure the switch for SSH connectivity.

Use the same commands that you used to configure SSH on the router in Part 2 to configure SSH for the switch.

- a. Configure the device name as listed in the Addressing Table.
- b. Configure the domain for the device.
`S1(config)# ip domain-name ccna-lab.com`
- c. Configure the encryption key method.
`S1(config)# crypto key generate rsa modulus 1024`
- d. Configure a local database username.
`S1(config)# username admin privilege 15 secret adminpass`
- e. Enable Telnet and SSH on the VTY lines.
`S1(config)# line vty 0 15`
`S1(config-line)# transport input telnet ssh`
- f. Change the login method to use the local database for user verification.
`S1(config-line)# login local`
`S1(config-line)# end`

Step 3: Establish an SSH connection to the switch.

Start Tera Term from PC-A, and then SSH to the SVI interface on S1.

Are you able to establish an SSH session with the switch?

Yes. SSH can be configured on a switch using the same commands that were used on the router.

Part 4: SSH From the CLI on the Switch

The SSH client is built into the Cisco IOS and can be run from the CLI. In Part 4, you will SSH to the router from the CLI on the switch.

Step 1: View the parameters available for the Cisco IOS SSH client.

Use the question mark (?) to display the parameter options available with the **ssh** command.

```
S1# ssh ?
  -c      Select encryption algorithm
  -l      Log in using this user name
  -m      Select HMAC algorithm
  -o      Specify options
  -p      Connect to this port
  -v      Specify SSH Protocol Version
  -vrf    Specify vrf name
  WORD    IP address or hostname of a remote system
```

Step 2: SSH to R1 from S1.

- a. You must use the **-l admin** option when you SSH to R1. This allows you to log in as user **admin**. When prompted, enter **adminpass** for the password.

```
S1# ssh -l admin 192.168.1.1
Password:
*****
Warning: Unauthorized Access is Prohibited!
*****
```

```
R1#
```

- b. You can return to S1 without closing the SSH session to R1 by pressing **Ctrl+Shift+6**. Release the **Ctrl+Shift+6** keys and press **x**. The switch privileged EXEC prompt displays.

```
R1#
```

```
S1#
```

- c. To return to the SSH session on R1, press Enter on a blank CLI line. You may need to press Enter a second time to see the router CLI prompt.

```
S1#
[Resuming connection 1 to 192.168.1.1 ... ]
```

```
R1#
```

- d. To end the SSH session on R1, type **exit** at the router prompt.

```
R1# exit
```

```
[Connection to 192.168.1.1 closed by foreign host]
```

```
S1#
```

What versions of SSH are supported from the CLI?

Answers may vary. This can be determined by using the **ssh -v ?** on the command line. The 2960 switch running IOS version 15.0(2) supports SSH v1 and V2.

```
S1# ssh -v ?
```

```
1 Protocol Version 1
```

```
2 Protocol Version 2
```

Reflection

How would you provide multiple users, each with their own username, access to a network device?

Answers may vary. You would add each user's username and password to the local database using the **username** command. It is also possible to use a RADIUS or TACACS server, but this has not been covered yet.

Router Interface Summary Table

Router Interface Summary				
Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.

Device Configs - Final

Router R1

```

service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
enable secret 4 06YFDUHH61wAE/kLkDq9BGho1QM5EnRtoyr8cHAUg.2
!
no aaa new-model
memory-size iomem 10
!
no ip domain lookup
ip domain name ccna-lab.com
ip cef
no ipv6 cef
multilink bundle-name authenticated
!
username admin privilege 15 secret 4 QHjxdsVkjtP7VxKIcPsLdTiMIvyLkyjT1HbmYxZigc
!

```

```
interface GigabitEthernet0/0
  no ip address
  shutdown
  duplex auto
  speed auto
!
interface GigabitEthernet0/1
  ip address 192.168.1.1 255.255.255.0
  duplex auto
  speed auto
!
interface Serial0/0/0
  no ip address
  shutdown
  clock rate 2000000
!
interface Serial0/0/1
  no ip address
  shutdown
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
control-plane
!
!
banner motd ^C
*****
  Unauthorized Access is Prohibited!
*****
^C
!
line con 0
  password 7 00071A150754
  login
line aux 0
line 2
  no activation-character
  no exec
  transport preferred none
  transport input all
  transport output pad telnet rlogin lapb-ta mop udptn v120 ssh
  stopbits 1
line vty 0
  password 7 110A1016141D
  login local
  transport input telnet ssh
```



```
line vty 1 4
login local
transport input telnet ssh
!
scheduler allocate 20000 1000
!
end
```

Switch S1

```
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname S1
!
boot-start-marker
boot-end-marker
!
enable secret 4 06YFDUHH61wAE/kLkDq9BGho1QM5EnRtoyr8cHAUg.2
!
username admin privilege 15 secret 4 QHjxdsVkjtoP7VxKIcPsLdTiMIvyLkyjT1HbmYxZigc
no aaa new-model
system mtu routing 1500
!
!
no ip domain-lookup
ip domain-name ccna-lab.com
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
```

```
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.11 255.255.255.0
!
 ip http server
 ip http secure-server
!
!
 banner motd ^C
 *****
  Unauthorized Access is Prohibited!
 *****
^C
!
```

```
line con 0
password 7 060506324F41
login
line vty 0 4
password 7 060506324F41
login local
transport input telnet ssh
line vty 5 15
password 7 00071A150754
login local
transport input telnet ssh
!
end
```